ref input, SimPoint estimates — small-8way

L1D 8 sets x 8, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	20.65%	47.95%	56.30%	20.67%	48.00%	56.39%
parest	1/1	0.16%	0.43%	0.62%	0.64%	1.19%	7.86%
deepsjeng	1/1	2.92%	9.17%	16.25%	8.11%	23.58%	49.72%
leela	1/1	0.38%	1.36%	1.80%	6.07%	15.90%	45.53%
namd	1/1	3.27%	8.21%	17.87%	3.27%	8.22%	17.90%
lbm	1/1	10.48%	32.28%	44.97%	10.48%	32.28%	44.97%
nab	1/1	5.20%	15.37%	34.02%	6.75%	18.74%	49.89%
xz_cld	1/1	3.38%	12.11%	22.88%	5.08%	17.35%	36.71%
xz_cpu2006docs	1/1	1.47%	5.78%	23.30%	2.03%	7.38%	37.90%
xz_combined	1/1	2.85%	10.66%	26.32%	4.17%	14.65%	42.32%
imagick	1/1	0.23%	0.12%	3.44%	5.33%	12.37%	46.11%
blender	1/1	0.38%	0.88%	3.23%	3.72%	6.96%	42.51%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.18%	55.24%
omnetpp	1/1	0.00%	0.00%	0.00%	12.75%	24.67%	48.13%
xalancbmk	1/1	0.13%	0.26%	0.80%	4.92%	8.32%	39.84%
x264_pass1	1/1	7.07%	15.02%	38.53%	10.18%	21.51%	55.90%
x264_pass2	1/1	6.25%	13.38%	40.16%	8.63%	18.23%	56.21%
x264_seek	1/1	6.41%	13.66%	41.07%	8.82%	18.56%	57.26%
gcc_pp_O3	1/1	1.23%	2.73%	4.48%	16.24%	33.66%	64.23%
gcc_pp_O2	1/1	1.25%	2.75%	4.47%	16.60%	34.40%	63.73%
gcc_smaller	1/1	1.09%	2.19%	4.61%	13.76%	25.29%	64.09%
gcc_ref32_O5	1/1	0.91%	1.84%	3.40%	15.53%	30.63%	59.73%
gcc_ref32_O3	1/1	0.68%	1.85%	1.06%	11.25%	27.66%	19.94%
mcf	1/1	1.40%	2.06%	7.74%	2.33%	3.60%	12.43%
perlbench_checkspam	1/1	7.98%	9.83%	29.42%	22.33%	36.24%	69.19%
perlbench_diffmail	1/1	1.96%	4.19%	4.64%	17.81%	31.25%	54.38%
perlbench_splitmail	1/1	4.16%	4.19%	15.82%	23.27%	35.82%	68.04%
bwaves_1	1/1	2.51%	6.24%	18.43%	2.51%	6.24%	18.43%
bwaves_2	1/1	2.59%	6.43%	18.99%	2.59%	6.43%	18.99%
bwaves_3	1/1	2.51%	6.23%	18.45%	2.51%	6.23%	18.45%
bwaves_4	1/1	2.56%	6.32%	19.44%	2.56%	6.32%	19.44%
exchange2	1/1	5.23%	10.55%	12.68%	6.08%	12.17%	14.84%
fotonik3d	1/1	3.36%	9.32%	9.25%	3.38%	9.37%	9.40%
roms	1/1	4.37%	13.83%	8.97%	4.44%	13.96%	9.48%
wrf	1/1	0.91%	3.38%	2.59%	1.52%	4.76%	7.57%
cam4	1/1	3.14%	11.02%	11.03%	4.99%	15.77%	21.23%
pooled		3.78%	9.06%	17.53%	7.49%	16.44%	39.26%
mean		3.31%	8.11%	15.78%	8.49%	18.33%	39.00%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.86%	5.46%	33.56%	4.62%	12.90%	64.24%	11.55%	9.11%
parest	done	9.94%	1.37%	10.52%	1.33%	4.88%	73.98%	5.39%	34.21%
deepsjeng	done	8.51%	1.85%	11.04%	7.30%	4.09%	6.50%	2.53%	14.37%
leela	done	4.20%	0.06%	4.76%	0.00%	2.73%	27.29%	0.22%	22.54%
namd	done	11.53%	2.43%	14.69%	1.26%	2.42%	27.58%	7.51%	16.05%
lbm	done	17.45%	0.00%	6.03%	0.10%	75.51%	97.87%	100.00%	14.02%
nab	done	4.13%	0.09%	5.81%	0.26%	0.42%	77.32%	12.94%	4.29%
xz_cld	done	10.24%	1.02%	15.13%	0.48%	2.25%	36.46%	0.05%	0.81%
xz_cpu2006docs	done	9.31%	1.55%	10.31%	1.37%	5.86%	40.00%	6.83%	9.28%
xz_combined	done	9.53%	1.13%	12.39%	0.71%	3.66%	24.91%	0.02%	2.35%
imagick	done	2.19%	3.59%	2.78%	0.01%	0.12%	100.00%	62.71%	45.85%
blender	done	7.97%	2.59%	8.75%	1.31%	3.31%	18.37%	1.24%	50.45%
povray	done	8.45%	7.25%	10.53%	18.27%	3.15%	49.55%	51.51%	64.06%
omnetpp	done	16.27%	0.08%	21.69%	1.19%	5.70%	28.12%	50.69%	3.26%
xalancbmk	done	16.84%	11.17%	19.39%	20.10%	4.47%	71.10%	6.45%	6.24%
x264_pass1	done	2.37%	0.10%	2.80%	0.99%	3.12%	29.18%	8.27%	15.71%
x264_pass2	done	1.83%	0.11%	2.22%	0.76%	1.97%	37.28%	1.30%	19.19%
x264_seek	done	1.87%	0.12%	2.29%	0.86%	1.91%	32.75%	1.36%	14.40%
gcc_pp_O3	done	8.28%	0.85%	11.02%	4.69%	3.48%	43.74%	8.74%	12.71%
gcc_pp_O2	done	8.50%	0.81%	11.31%	4.47%	3.73%	39.97%	8.88%	11.39%
gcc_smaller	done	10.11%	1.46%	12.99%	3.25%	3.32%	49.17%	4.79%	11.57%
gcc_ref32_O5	done	8.29%	1.16%	10.85%	3.63%	3.63%	53.60%	10.84%	14.30%
gcc_ref32_O3	done	10.39%	1.97%	11.75%	5.03%	9.58%	74.28%	4.96%	18.32%
mcf	done	21.44%	5.22%	27.38%	6.19%	4.64%	74.48%	4.06%	20.04%
perlbench_checkspam	done	1.96%	0.09%	3.07%	0.39%	0.87%	34.15%	2.72%	9.90%
perlbench_diffmail	done	4.31%	0.12%	5.90%	6.10%	1.62%	39.09%	0.13%	9.08%
perlbench_splitmail	done	8.75%	0.03%	12.45%	6.70%	2.89%	8.56%	2.96%	0.45%
bwaves_1	done	24.64%	6.55%	25.37%	0.07%	35.29%	34.83%	2.78%	0.83%
bwaves_2	done	24.25%	6.28%	24.90%	0.08%	36.22%	35.39%	2.67%	0.95%
bwaves_3	done	21.04%	10.96%	20.59%	0.00%	37.89%	40.82%	1.81%	94.33%
bwaves_4	done	19.67%	10.48%	19.12%	0.00%	37.67%	43.40%	1.23%	53.65%
exchange2	done	0.96%	1.14%	1.29%	1.80%	0.37%	1.61%	0.12%	0.50%
fotonik3d	done	8.47%	2.73%	11.86%	0.03%	0.02%	81.02%	2.80%	11.65%
roms	done	11.98%	1.78%	14.08%	0.61%	11.40%	67.67%	17.23%	11.63%
wrf	done	6.51%	5.81%	6.48%	4.04%	6.78%	57.32%	9.71%	12.77%
cam4	done	10.43%	1.79%	11.24%	2.13%	12.02%	56.87%	9.17%	27.49%
pooled		9.07%	3.08%	10.99%	2.72%	6.13%	54.37%	9.99%	11.71%
mean		10.24%	2.76%	12.12%	3.06%	9.72%	46.62%	11.84%	18.55%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.